

An Introduction to Quantum Computing

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Abstract. These lecture notes introduce quantum computing to graduate students in mathematics with a background in numerical linear algebra and numerical analysis. The central theme is *Quantum Phase Estimation* (QPE): every major algorithm — Shor’s factoring algorithm, Grover search and amplitude estimation, the HHL quantum linear solver, and continuous-time quantum walks — is an instance of QPE applied to a carefully chosen unitary, with Stone’s theorem connecting each unitary to its Hermitian generator. We develop the necessary foundations (quantum states, gates, measurement), the Quantum Fourier Transform as the computational engine, and QPE as a quantum eigenvalue algorithm. We then treat Shor’s algorithm via the eigenstructure of modular exponentiation, Grover’s algorithm and amplitude estimation, Hamiltonian simulation via product formulas, HHL, and continuous-time quantum walks. The course concludes with a survey of the quantum advantage landscape — distinguishing unconditional, hardness-conditional, and input-model-conditional speedups — a discussion of dequantization, and four open problems at the interface of quantum computing and numerical analysis. Throughout, connections to classical numerical methods are emphasized.

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1 Foundations

Quantum computing is, at its core, linear algebra over complex vector spaces, augmented by the physical constraints of quantum mechanics. We present the mathematical framework as four axioms, requiring no background in physics.

Why should we care? The central promise of quantum computing is computational speedup for problems that are intractable classically. For numerical mathematicians, the most relevant examples are: solving linear systems $Ax = b$ exponentially faster in the system size N ; computing matrix functions such as e^A efficiently for large sparse matrices A ; and estimating integrals and expectations with a quadratic improvement over Monte Carlo. These speedups come with important caveats — they require a quantum computer that does not yet exist at scale, and the results are often quantum states rather than classical vectors. Nevertheless, the algorithms are built from ideas that are deeply familiar to classical numerical methods: eigenvalue methods, Fourier analysis, operator splitting, and rational approximation. These notes try to develop those connections throughout.

The four postulates

State space. The *state* of a quantum system is described by a unit vector $|\psi\rangle$ in an N -dimensional complex Hilbert space $\mathcal{H}$. For n *qubits*, $\mathcal{H} = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$.

Evolution. Closed-system evolution is described by a *unitary operator* $U \in \mathcal{U}(\mathcal{H})$. The state $|\psi\rangle$ becomes $U|\psi\rangle$ after applying the unitary U .

Measurement. A *projective measurement* with respect to an orthonormal basis $\{|k\rangle\}_{k=0}^{N-1}$ on state $|\psi\rangle$ yields outcome k with probability

$$p_k = |\langle k | \psi \rangle|^2, \quad (\text{Born's rule})$$

and leaves the post-measurement state $|k\rangle$.

Composition. The joint state space of systems A and B is $\mathcal{H}_A \otimes \mathcal{H}_B$. A gate U_A acting on A alone acts as $U_A \otimes \mathbb{I}_B$ on the joint system.

Remark 1.1. Each postulate has a direct interpretation in numerical linear algebra. A quantum state is a unit vector in $\mathbb{C}^{2^n}$, whose dimension grows exponentially in n . Gates are unitary matrices, so they are norm-preserving. The Born rule $p_k = |\langle k | \psi \rangle|^2$ is the squared magnitude of a projection coefficient in an orthonormal basis expansion. The tensor product $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n} \cong \mathbb{C}^{2^{m+n}}$ is the standard multilinear tensor product.

Two states differing only by a *global phase*, $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ with $\gamma \in \mathbb{R}$, are indistinguishable: for every measurement basis $\{|k\rangle\}$ the Born probabilities $|\langle k | e^{i\gamma}\psi \rangle|^2 = |\langle k | \psi \rangle|^2$ coincide, and this persists after any later unitary (which commutes with the scalar $e^{i\gamma}$).

Using this freedom to make the $|0\rangle$ -amplitude real and non-negative, every single-qubit state can be written

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi).$$

The angles (θ, ϕ) are the spherical coordinates of a point on the unit sphere $S^2 \subset \mathbb{R}^3$, the *Bloch sphere*: $|0\rangle, |1\rangle$ are the north and south poles, while $|\pm\rangle$ and $|\pm y\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$ lie on the equator. Antipodal points are precisely the orthogonal states. Single-qubit unitaries act as rigid rotations of this sphere: $R_x(\alpha)$, $R_y(\alpha)$, $R_z(\alpha)$ rotate by α about the $\hat{x}$, $\hat{y}$, $\hat{z}$ axes, and every $U \in \mathcal{U}(2)$ is, up to global phase, such a rotation (Euler decomposition $U = e^{i\gamma}R_z(\beta)R_y(\delta)R_z(\lambda)$).

Exercise 1.2. Let $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle \in \mathbb{C}^2$ be an arbitrary single-qubit state.

- (a) Compute the probability of outcome $|0\rangle$ when measuring in the computational basis.
- (b) Compute the probability of outcome $|+\rangle$ when measuring in the basis $\{|+\rangle, |-\rangle\}$, where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

- (c) For which θ and ϕ are the two outcomes equally likely in both bases simultaneously?

Qubits and Dirac notation

A *qubit* is a unit vector in $\mathbb{C}^2$. The *computational basis* is $|0\rangle = (1, 0)^\top$ and $|1\rangle = (0, 1)^\top$, so a general qubit state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, $\alpha, \beta \in \mathbb{C}$.

For n qubits, the computational basis is $\{|x\rangle : x \in \{0, 1\}^n\}$, where $|x_1 \cdots x_n\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle$. A general n -qubit state is $|\psi\rangle = \sum_{x \in \{0, 1\}^n} \alpha_x |x\rangle$ with $\sum_x |\alpha_x|^2 = 1$.

We use Dirac notation throughout. For $|\psi\rangle, |\phi\rangle \in \mathcal{H}$:

- $\langle\psi| = |\psi\rangle^\dagger$ is the conjugate transpose (row vector);
- $\langle\phi|\psi\rangle \in \mathbb{C}$ is the inner product;
- $|\phi\rangle\langle\psi|$ is the rank-1 operator (outer product);
- $\mathbb{I} = \sum_k |k\rangle\langle k|$ is the identity (resolution of the identity).

Standard quantum gates

Quantum gates are unitary matrices. The single-qubit *Pauli gates* are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The *rotation gates* around the three Pauli axes are defined by

$$R_x(\theta) = e^{-i(\theta/2)X} = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix},$$

$$R_y(\theta) = e^{-i(\theta/2)Y} = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \quad R_z(\theta) = e^{-i(\theta/2)Z} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

The *Hadamard gate* and *phase gates* are

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \quad R_\varphi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}.$$

Note that $P \in \{X, Y, Z, H\}$ are Hermitian ($P^\dagger = P$).

The canonical two-qubit gate is the *controlled-NOT*:

$$\text{CNOT} = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \text{CNOT}|c\rangle|t\rangle = |c\rangle|c \oplus t\rangle.$$

More generally, a *controlled-U* gate applies U to the target qubit iff the control qubit is $|1\rangle$:

$$\text{c-}U = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U.$$

Exercise 1.3. (a) Determine the eigenvalues and eigenvectors of X and Z .

(b) Show that H is the change-of-basis matrix between the eigenbases of Z and X , and verify the identity $HZH = X$ from this.

Exercise 1.4. Show that each Pauli gate satisfies $P^2 = \mathbb{I}$ for $P \in \{X, Y, Z, H\}$. Conclude from this that P takes the form $P = e^{i(\pi/2)(\mathbb{I}-P)}$.

The Hadamard transform

Proposition 1.5 (Hadamard transform). *For any $x \in \{0, 1\}^n$,*

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle,$$

where $x \cdot z = \sum_i x_i z_i \bmod 2$. In particular, $H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_z |z\rangle$ is the uniform superposition over all basis states.

Proof. For $n = 1$: $H|x_1\rangle = \frac{1}{\sqrt{2}} \sum_{z_1} (-1)^{x_1 z_1} |z_1\rangle$. For general n , apply the base case qubit-by-qubit and collect the signs over the tensor product:

$$H^{\otimes n}|x\rangle = \bigotimes_{i=1}^n H|x_i\rangle = \bigotimes_{i=1}^n \frac{1}{\sqrt{2}} \sum_{z_i} (-1)^{x_i z_i} |z_i\rangle = \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle. \quad \square$$

The Hadamard transform requires only n single-qubit gates (one H per qubit), compared to $O(n \cdot 2^n)$ operations classically. This exponential compression is possible because it acts on amplitudes in superposition, not on 2^n data entries individually. It is the prototype for the *Quantum Fourier Transform* in Section 2.

The key structural difference from classical computation is this: a classical algorithm that applies the Hadamard transform to a vector $v \in \mathbb{C}^{2^n}$ must read all 2^n entries of v . The quantum algorithm applies $H^{\otimes n}$ to a quantum state in $O(n)$ gates, but the result is a *superposition*, not a list of coefficients. Extracting individual coefficients still requires measurement, which collapses the superposition. This tension between the exponential state space and the limited classical information one can extract from it is the real challenge in designing useful quantum algorithms.

Entanglement

Definition 1.6 (Entanglement). A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *separable* if $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$ for some $|\phi\rangle \in \mathcal{H}_A$, $|\chi\rangle \in \mathcal{H}_B$; otherwise it is *entangled*.

The four *Bell states* form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$, all maximally entangled:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

Proposition 1.7 (Bell state preparation). $\text{CNOT} \cdot (H \otimes \mathbb{I})|00\rangle = |\Phi^+\rangle$.

Exercise 1.8. Verify Proposition 1.7. Then construct circuits (sequences of H , X , and CNOT gates) that prepare each of $|\Phi^-\rangle$, $|\Psi^+\rangle$, and $|\Psi^-\rangle$ from the input $|00\rangle$.

Remark 1.9. A separable state $|\phi\rangle \otimes |\chi\rangle$ corresponds to a rank-1 tensor in $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n}$. When a general state is viewed as a matrix $M \in \mathbb{C}^{2^m \times 2^n}$, with M_{ij} the coefficient of $|ij\rangle$, the state is separable if and only if $\text{rank}(M) = 1$. The singular value decomposition of M yields the *Schmidt decomposition*, and the number of non-zero singular values is the *Schmidt rank*, which quantifies the degree of entanglement.

Exercise 1.10. (a) Show that $|\Phi^+\rangle$ has Schmidt rank 2.

(b) For the state $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, find the Schmidt decomposition and determine whether it is entangled.

The following is a very important physical restriction in quantum computing. It is also one of the main reasons quantum computing is difficult.

Proposition 1.11 (No-cloning theorem [1]). *There is no unitary U satisfying*

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \text{for all } |\psi\rangle \in \mathbb{C}^2.$$

Proof. Suppose such U exists. For any $|\phi\rangle, |\psi\rangle$, unitarity gives

$$\langle\phi|\psi\rangle^2 = (\langle\phi|\otimes\langle 0|)(U^\dagger U)(|0\rangle\otimes|\psi\rangle) = \langle\phi|\psi\rangle$$

so $\langle\phi|\psi\rangle \in \{0, 1\}$ for all pairs. This contradicts the existence of non-orthogonal, non-identical states such as $|0\rangle$ and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. $\square$

Reversible computation and oracles

Unitaries are invertible ($U^{-1} = U^\dagger$), so quantum computation is *reversible*. On the other hand, an irreversible classical map must be embedded reversibly before it can be run. A way to do this is the following: For $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ the standard embedding is the (*XOR*) *oracle*

$$U_f |x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle, \quad x \in \{0, 1\}^n, y \in \{0, 1\}^m,$$

which gives a permutation of basis states. Hence, U_f is unitary on $\mathbb{C}^n \otimes \mathbb{C}^m$ and is an involution ($U_f^2 = \mathbb{I}$). The input register x is preserved; the answer is XORed into the second register. Recall that $\oplus$ is the sum modulo 2, i.e., the XOR gate is a classical logic gate that outputs 0 when the components of y and $f(x)$ are equal, and 1 otherwise.

Phase kickback. Setting the target register to the X-eigenstate $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ turns the XOR oracle into a *phase oracle*. Indeed, since $X|-\rangle = -|-\rangle$, it holds that

$$U_f |x\rangle|-\rangle = |x\rangle \frac{1}{\sqrt{2}}(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle) = (-1)^{f(x)} |x\rangle|-\rangle.$$

The function value is “kicked back” onto the input register as a relative phase $(-1)^{f(x)}$, while $|-\rangle$ is left untouched. More generally, a controlled- U whose target is an eigenstate $|u\rangle$, e.g., $U|u\rangle = e^{2\pi i \varphi}|u\rangle$ imprints the eigenphase on the control:

$$\text{c-}U |1\rangle|u\rangle = e^{2\pi i \varphi} |1\rangle|u\rangle.$$

Converting eigenvalue information into a measurable relative phase is the single mechanism behind quantum phase estimation (Section 3) and, hence, behind every algorithm on the QPE spine, such as Grover (Section 5).

2 Quantum Fourier Transform

The *Quantum Fourier Transform* is the computational engine of the algorithms in this course. It is the DFT matrix acting on amplitudes of a quantum state, and it admits an efficient $O(n^2)$ -gate circuit arising from a product formula.

Why the Fourier transform? The DFT is ubiquitous in numerical analysis: it converts between a function and its frequency representation, diagonalizes circulant matrices, and accelerates convolution from $O(N^2)$ to $O(N \log N)$ via the FFT. In quantum computing, the DFT plays an analogous role: it converts between a computational-basis representation and a phase-encoded representation, and it is the tool by which quantum algorithms *read off* periodicity and spectral information from a quantum state. The key difference is that the QFT operates on the *amplitudes* of a superposition — it is applied to 2^n complex numbers simultaneously in $O(n^2)$ gate operations, a task that would take $O(n \cdot 2^n)$ classical operations. As with the Hadamard transform, this compression is only useful when the circuit's output state is processed by a subsequent algorithm (such as QPE) that extracts the relevant information without full tomography.

Definition and unitarity

Throughout this section, $n \in \mathbb{N}$ denotes the number of qubits and $N = 2^n$. We use three related but distinct objects:

- an *integer* $k \in \{0, \dots, N-1\}$, written in decimal as usual;
- its *binary representation* $k = k_1 2^{n-1} + k_2 2^{n-2} + \dots + k_n 2^0$, written as the bit string $[k_1 k_2 \dots k_n]$ with each $k_j \in \{0, 1\}$;
- the *binary fraction* $[0.k_l k_{l+1} \dots k_m] = k_l 2^{-1} + \dots + k_m 2^{-(m-l+1)} \in [0, 1)$.

The computational basis state $|k\rangle$ is indexed by the *integer* $k \in \mathbb{N}$; we write $|k_1 k_2 \dots k_n\rangle$ to mean the tensor product $|k_1\rangle \otimes |k_2\rangle \otimes \dots \otimes |k_n\rangle$, which equals $|k\rangle$ when $k = \sum_j k_j 2^{n-j}$. Binary fractions will commonly appear in exponents as phases, e.g., $e^{2\pi i [0.k_1 k_2]}$.

The *Discrete Fourier Transform* (DFT) on $\mathbb{C}^N$ is the linear map F_N defined by

$$\hat{a}_k := (F_N a)_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega_N^{jk} a_j, \quad k = 0, \dots, N-1, \quad \omega_N := e^{2\pi i/N}.$$

with matrix elements $(F_N)_{kj} = \omega_N^{jk}/\sqrt{N}$, $j, k = 0, \dots, N-1$.

Lemma 2.1 (Unitarity of the DFT). $F_N^\dagger F_N = \mathbb{I}_N$.

Proof. $(F_N^\dagger F_N)_{lj} = \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{k(j-l)}$. For $j = l$ this equals 1. For $j \neq l$, the sum is a geometric series with ratio $\omega_N^{j-l} \neq 1$, which vanishes. $\square$

Definition 2.2 (Quantum Fourier Transform). The n -qubit *Quantum Fourier Transform* QFT_N ($N = 2^n$) is the unitary operator defined on the computational basis by

$$\text{QFT}_N |k\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega_N^{jk} |j\rangle, \quad k = 0, \dots, N-1.$$

As a matrix, $\text{QFT}_N = F_N$, which is unitary by Lemma 2.1.

Remark 2.3. The QFT is the DFT matrix F_N applied to the *amplitude vector* of a quantum state rather than to a classical data vector. A classical computer requires $O(N \log N)$ arithmetic operations (FFT) to apply F_N to an N -vector. The QFT circuit uses only $O(n^2) = O((\log N)^2)$ gates, but it acts on a quantum superposition of all N basis states simultaneously. This apparent compression is why the QFT is the engine of Shor's algorithm and QPE, though extracting the result requires careful design of the surrounding algorithm.

From the definition of the QFT, it is not apparent that it is a transformation on n qubits. The following statement makes it explicit.

Theorem 2.4 (Product formula). *The quantum Fourier transform takes the form*

$$\text{QFT}_N |k_1 k_2 \cdots k_n\rangle = \bigotimes_{l=1}^n \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i [0.k_{n-l+1} \cdots k_n]} |1\rangle \right).$$

Proof. Substituting the binary expansion of k and $|j\rangle = |j_1\rangle \otimes \cdots \otimes |j_n\rangle$:

$$\begin{aligned} \text{QFT}_N |k\rangle &= \frac{1}{\sqrt{N}} \sum_{j_1, \dots, j_n \in \{0,1\}} e^{2\pi i k \sum_{l=1}^n j_l 2^{-l}} |j_1\rangle \otimes \cdots \otimes |j_n\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{j_1, \dots, j_n \in \{0,1\}} \prod_{l=1}^n e^{2\pi i k \cdot 2^{-l} j_l} |j_1\rangle \otimes \cdots \otimes |j_n\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{j_1, \dots, j_n \in \{0,1\}} \left(e^{2\pi i k \cdot 2^{-1} j_1} |j_1\rangle \right) \otimes \cdots \otimes \left(e^{2\pi i k \cdot 2^{-n} j_n} |j_n\rangle \right) \\ &= \frac{1}{\sqrt{N}} \bigotimes_{l=1}^n \left(|0\rangle + e^{2\pi i k \cdot 2^{-l}} |1\rangle \right). \end{aligned}$$

Writing $k = k_1 2^{n-1} + \cdots + k_n$, we have $k \cdot 2^{-l} = k_1 2^{n-1-l} + \cdots + k_{n-l} + [0.k_{n-l+1} \cdots k_n]$. The integer part contributes only multiples of 2π to the exponent, so $e^{2\pi i k \cdot 2^{-l}} = e^{2\pi i [0.k_{n-l+1} \cdots k_n]}$. $\square$

Remark 2.5. The product formula reveals two key structural facts. First, the output is a *tensor product* of single-qubit states, enabling a qubit-by-qubit circuit. Second, the l -th factor depends only on the last l bits $k_{n-l+1}, \dots, k_n$ of the input, so the output qubits are in *reversed* order relative to the input.

Exercise 2.6. Show that QFT_N diagonalizes the cyclic shift operator $\hat{X}|k\rangle = |k+1 \bmod N\rangle$, i.e., compute the eigenvectors and eigenvalues of $\hat{X}$, and verify that

$$\text{QFT}_N^\dagger \hat{X} \text{QFT}_N = \hat{Z}^{-1},$$

where $\hat{Z}|k\rangle = \omega_N^k |k\rangle$.

Interpret this as the quantum analog of the DFT diagonalizing the derivative operator.

Exercise 2.7. Show that $\text{QFT}_N^4 = \mathbb{I}$ for all $N \in \mathbb{N}$, and deduce that all the eigenvalues of QFT_N lie in the set $\{1, i, -1, -i\}$.

Exercise 2.8. Apply QFT_4 to the states $|1\rangle$ and $|2\rangle$ using the product formula (cf. Theorem 2.4). Express each result as a tensor product of single-qubit states and verify using the definition that

$$\text{QFT}_4 |k\rangle = \frac{1}{2} \sum_{j=0}^3 i^{jk} |j\rangle.$$

2.1 The QFT circuit

The product formula directly yields an efficient circuit. Define the *phase rotation gate*

$$R_m = \text{diag}(1, e^{2\pi i/2^m}), \quad m \in \mathbb{N},$$

and the *controlled- R_m* gate, which applies R_m to the target iff the control is $|1\rangle$, i.e.,

$$\text{c-}R_m = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes R_m.$$

Theorem 2.9 (QFT circuit complexity [1]). *QFT $_N$ can be implemented using n Hadamard gates, $\binom{n}{2}$ controlled- R_m gates, and $\lfloor n/2 \rfloor$ SWAP gates. Total gate count: $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor = O(n^2)$.*

Example 2.10 ($n = 3$ qubits). In Figure 2.1, we trace the circuit for input $|k\rangle = |k_1 k_2 k_3\rangle$.

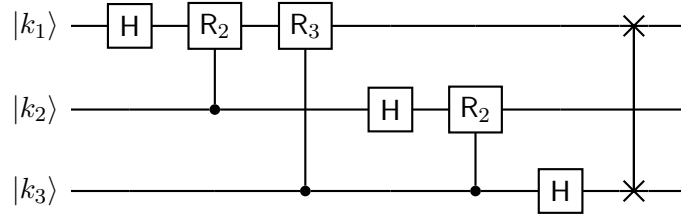


Figure 2.1: QFT circuit for $n = 3$ qubits. After the final SWAP the output order matches the product formula.

We begin by tracing what happens to qubit 1 (the most significant):

$$(1) \text{ H on qubit 1: } |k_1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{k_1}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_1]}|1\rangle).$$

$$(2) \text{ Controlled-}R_2 \text{ (control: qubit 2): } \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_1 k_2]}|1\rangle).$$

$$\begin{aligned} |0\rangle \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_1]}|1\rangle) &\rightarrow |0\rangle \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (k_1 2^{-1} + k_2 2^{-2})}|1\rangle) \\ |1\rangle \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_1]}|1\rangle) &\rightarrow |1\rangle \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (k_1 2^{-1} + k_2 2^{-2})}|1\rangle) \end{aligned}$$

$$(3) \text{ Controlled-}R_3 \text{ (control: qubit 3): } \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_1 k_2 k_3]}|1\rangle).$$

Each controlled- R_m appends one further binary digit to the phase. The same procedure on qubit 2 yields $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_2 k_3]}|1\rangle)$, and H alone on qubit 3 gives $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i [0.k_3]}|1\rangle)$. After the SWAP, the output matches Theorem 2.4. Gate count for $n = 3$: $3 + 3 + 1 = 7$, consistent with $\frac{3 \cdot 4}{2} + 1 = 7$. For general n : qubit l receives H plus $(n - l)$ controlled rotations, giving $\sum_{l=1}^n (n - l) = \binom{n}{2}$ controlled- R_m gates, n Hadamards, and $\lfloor n/2 \rfloor$ SWAPs. $\square$

Since QFT $_N$ is unitary, QFT $_N^{-1} = \text{QFT}_N^\dagger$. As usual, the inverse QFT serves as a basis change, transforming from the Fourier basis to the computational basis.

Proposition 2.11 (Inverse QFT circuit). *QFT $_N^\dagger$ is implemented by reversing the QFT circuit and replacing each R_m with $R_m^\dagger = R_{-m}$ (phase $e^{-2\pi i/2^m}$).*

Remark 2.12. The gates R_m apply exponentially small rotations $e^{2\pi i/2^m}$ for $m \gg 1$ large and may be omitted in practice; see Section 2.2.

2.2 Approximate QFT

Definition 2.13 (Approximate QFT). The *approximate QFT* with cutoff $m_{\max}$ is obtained by dropping all controlled- R_m gates with $m > m_{\max}$. It uses $\Theta(n m_{\max})$ gates instead of $\Theta(n^2)$.

Theorem 2.14 (Approximation error). Let U and U_{approx} be the exact and approximate QFT unitaries. Then

$$\|U - U_{\text{approx}}\|_{\text{op}} \leq \frac{2\pi n}{2^{m_{\max}}},$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. Consequently, choosing $m_{\max} = \lceil \log_2(n/\varepsilon) \rceil$ yields an error less than $\varepsilon > 0$ with only $O(n \log(n/\varepsilon))$ gates.

Proof. Write $U = G_1 G_2 \cdots G_K$ and $U_{\text{approx}} = G'_1 \cdots G'_K$, where $G'_i = G_i$ for kept gates and $G'_i = \mathbb{I}$ for dropped gates. By the triangle inequality applied inductively to products of unitaries,

$$\|U - U_{\text{approx}}\|_{\text{op}} \leq \sum_{i: G_i \text{ dropped}} \|G_i - \mathbb{I}\|_{\text{op}}.$$

A $c\text{-}R_m$ gate in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is $\text{diag}(1, 1, 1, e^{2\pi i/2^m})$. Hence,

$$\|c\text{-}R_m - \mathbb{I}\|_{\text{op}} = |e^{2\pi i/2^m} - 1| = 2 \sin\left(\frac{\pi}{2^m}\right) \leq \frac{2\pi}{2^m},$$

where we used the estimates $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$ and $|\sin \theta| \leq |\theta|$. Summing over all dropped gates (those with $m > m_{\max}$):

$$\|U - U_{\text{approx}}\|_{\text{op}} \leq \sum_{l=1}^n \sum_{m=m_{\max}+1}^{n-l+1} \frac{2\pi}{2^m} \leq n \sum_{m=m_{\max}+1}^{\infty} \frac{2\pi}{2^m} = \frac{2\pi n}{2^{m_{\max}}}. \quad \square$$

Remark 2.15. The approximation strategy mirrors two classical ideas. First, it is analogous to a *low-pass filter*: rotations R_m with $m > m_{\max}$ correspond to high-frequency components $e^{2\pi i/2^m}$ near 1, which contribute negligibly to the output and can be discarded. Second, the gate count $O(n \log(n/\varepsilon))$ matches the complexity of computing a *sparse* DFT (keeping only the $O(n m_{\max})$ most significant frequency interactions), and is reminiscent of the complexity of quadrature rules with adaptive frequency truncation. The Frobenius error bound plays the role of the aliasing error estimate in such truncated transforms.

Exercise 2.16. For $n = 4$ qubits and cutoff $m_{\max} = 2$, identify which controlled- R_m gates are dropped. Compute the Frobenius error bound from Theorem 2.14 and compare with the total number of gates saved.

3 Quantum Phase Estimation

Quantum Phase Estimation (QPE) [1, 3] is the central algorithmic primitive of this course. Every subsequent algorithm — Shor, HHL, and amplitude estimation — uses QPE as a subroutine applied to a carefully chosen unitary.

Motivation. The QPE problem is to estimate the eigenphase φ of a unitary $U \in \mathcal{U}(\mathbb{C}^d)$ given an eigenvector $|u\rangle \in \mathbb{C}^d$, i.e., $U|u\rangle = e^{2\pi i\varphi}|u\rangle$. A natural classical approach is to observe the scalar sequence

$$s_k := \langle u, U^k u \rangle = e^{2\pi i k \varphi}, \quad k = 0, 1, \dots, N \in \mathbb{N}.$$

This is a pure complex tone at frequency φ , sampled at integer times $k \in \mathbb{N}$. Its DFT is

$$\hat{s}_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} s_k e^{-2\pi i k j / N} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i k (\varphi - j/N)},$$

a geometric series concentrated at $j = \lfloor N\varphi \rfloor$. The eigenphase φ is therefore estimated by finding the peak of $|\hat{s}_j|^2$ — a standard frequency estimation procedure — which, like in FFT, requires $O(N \log N)$ arithmetic operations.

QPE implements this procedure differently on a quantum computer, outlined in Algorithm 3.1 — see Figure 3.1 for the circuit diagram.

Setup. Let $U \in \mathcal{U}(\mathbb{C}^d)$ be a system unitary operator on $\mathbb{C}^d$ with eigenpair $(e^{2\pi i\varphi}, |u\rangle)$, where $\varphi \in [0, 1)$ is the unknown eigenphase to be determined. QPE uses a t -qubit *clock register* starting with $|0^t\rangle$ and the *system register* holding $|u\rangle$ to estimate φ to $O(t)$ bits of precision.

The clock register is brought into uniform superposition by applying $H^{\otimes t}$, i.e.,

$$H^{\otimes t} \otimes \mathbb{I} : |0^t\rangle|u\rangle \mapsto \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle|u\rangle.$$

Equivalently, applying QFT_{2^t} to $|0^t\rangle$.

Phase kickback. A key mechanism in QPE is *phase kickback*: when a controlled- U gate is applied with the clock qubit in superposition and the system in eigenstate $|u\rangle$, we get

$$\text{controlled-}U : \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\varphi}|1\rangle)|u\rangle.$$

The eigenphase $e^{2\pi i\varphi}$ is *kicked back* onto the clock qubit while the system register is left unchanged. Similarly, applying controlled- U^{2^l} with the $(l+1)$ -th clock qubit kicks back $e^{2\pi i 2^l \varphi}$. Doing this for all t clock qubits builds up the binary fraction representation of an approximation of φ in the clock register, i.e., we obtain in the clock register the state

$$\bigotimes_{l=1}^t \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 2^{l-1} \varphi} |1\rangle \right) = \bigotimes_{l=1}^t \frac{1}{\sqrt{2}} \sum_{k_l \in \{0,1\}} e^{2\pi i k_l 2^{l-1} \varphi} |k_l\rangle = \frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i k \varphi} |k\rangle.$$

Readout. Finally, applying $\text{QFT}_t^\dagger$ to the clock concentrates amplitude at the integer $b = \lfloor 2^t \varphi \rfloor$ nearest to $2^t \varphi$, from which the estimate $\tilde{\varphi} = b/2^t$ is read off by measurement. In particular,

$$\begin{aligned} \text{QFT}_t^\dagger \left(\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i k \varphi} |k\rangle \right) &= \text{QFT}_t^\dagger \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{k \cdot N \varphi} |k\rangle \right), \quad N = 2^t \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{k \cdot N \varphi} \sum_{j=0}^{N-1} \omega_N^{-kj} |j\rangle = \sum_{j=0}^{N-1} \left(\frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{k(N\varphi-j)} \right) |j\rangle. \end{aligned}$$

If $j \neq N\varphi$, then the amplitude α_j of the basis state $|j\rangle$ is given by

$$\alpha_j = \frac{1}{N} \sum_{k=0}^{N-1} \omega_N^{k(N\varphi-j)} = \frac{1}{N} \frac{1 - e^{2\pi i k(N\varphi-j)}}{1 - e^{2\pi i k(\varphi-j/N)}},$$

which concentrates on $b = \lfloor 2^t \varphi \rfloor$. Hence, the QPE map gives $\text{QPE}(|0^t\rangle|u\rangle) \approx |2^t \tilde{\varphi}\rangle|u\rangle$.

Measurement. QPE ends with a single measurement of the t -qubit clock register, yielding an integer $k \in \{0, \dots, 2^t - 1\}$ and costing $O(t)$ classical bits. If φ is exactly representable in t bits (i.e., $2^t \varphi \in \mathbb{Z}$), the outcome is deterministic. Otherwise, the outcome is random, and the nearest t -bit approximation to φ is obtained with probability at least $4/\pi^2 \approx 0.405$ when using exactly $t = n$ clock qubits. There are two strategies to boost this to $1 - \epsilon$:

- (i) add $\lceil \log_2(2 + 1/(2\epsilon)) \rceil$ extra clock qubits, which Theorem 3.2 below shows achieves success probability $\geq 1 - \epsilon$ in a single run, or
- (ii) use $t = n$ qubits, repeat $O(\log(1/\epsilon))$ times, and take the median (*median amplification*).

Strategy (i) uses more qubits; strategy (ii) uses more circuit executions.

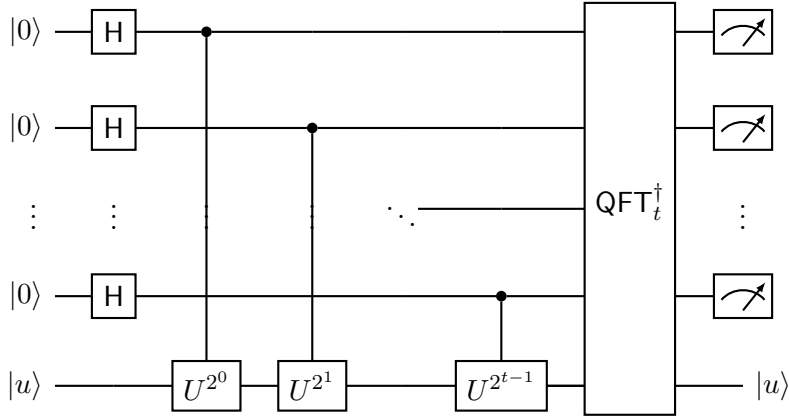


Figure 3.1: QPE circuit with t clock qubits (top) and system register (bottom). The Hadamard layer creates a uniform superposition in the clock. Controlled- U^{2^j} gates accumulate the eigenphase φ in the clock register. The inverse QFT then converts the encoded phase into a computational-basis label.

Remark 3.1. QPE requires applying the controlled- U^{2^j} gates for $j = 0, \dots, t - 1$. If U is given as a circuit of g gates, then U^{2^j} costs $O(g2^j)$ gates by repetition, and the total over all j is

Algorithm 3.1 Quantum Phase Estimation

Input: Unitary U with eigenstate $|u\rangle$; $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ clock qubits

Output: Estimate $\tilde{\varphi} = j/2^t$ with $|\tilde{\varphi} - \varphi| \leq 2^{-n}$ with probability $\geq 1 - \epsilon$

- 1: Initialise clock and system in $|0\rangle^{\otimes t}|u\rangle$
 - 2: Apply $H^{\otimes t}$ to clock: $\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle|u\rangle$
 - 3: **for** $j = 0, 1, \dots, t-1$ **do**
 - 4: Apply controlled- U^{2^j} with clock qubit $j+1$ as control
 - 5: **end for** $\triangleright$ Phase kickback gives clock state $\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i j \varphi} |j\rangle$
 - 6: Apply $\text{QFT}_t^\dagger$ to clock register
 - 7: **Measure** clock register; **return** j
-

$O(g 2^t)$ — exponential in t , which defeats the purpose. QPE is therefore efficient only when the structure of U allows U^{2^j} to be implemented at polynomial cost. In this course, two such instances arise: Shor's algorithm (Section 4) and HHL (Section 7).

This is entirely analogous to the classical situation: computing $A^{2^j}v$ by repeated matrix-vector products costs $O(2^j)$ matvecs without structure. In both classical and quantum eigenvalue algorithms, efficiency requires exploiting the specific form of the operator.

Theorem 3.2 (QPE precision). *Let $\epsilon > 0$ and consider $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ clock qubits. Then QPE outputs a t -bit approximation $\tilde{\varphi}$ with $|\tilde{\varphi} - \varphi| \leq 2^{-n}$ and probability at least $1 - \epsilon$.*

Proof. Below, we set $N = 2^t$ and use the elementary inequality

$$|1 - e^{i\theta}| \geq \frac{2|\theta|}{\pi} \quad \text{for } \theta \in [-\pi, \pi]. \quad (3.1)$$

Now let $b \in \{0, \dots, N-1\}$ be the integer with $b \leq N\varphi < b+1$, and set $\delta := \varphi - b/N \in [0, \frac{1}{N})$.

After applying $\text{QFT}_t^\dagger$, the amplitude of outcome for the basis state $|j\rangle$ is given by

$$\alpha_j = \frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i k(\varphi - j/N)} = \frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i k(\delta - (j-b)/N)}.$$

Amplitude bound. For $j \neq b$, the ratio is not 1, and the geometric sum gives

$$\alpha_j = \frac{1}{N} \frac{1 - e^{2\pi i(N\delta - (j-b))}}{1 - e^{2\pi i(\delta - (j-b)/N)}}.$$

Moreover, we have that $\delta - (j-b)/N \in [-\frac{1}{2}, \frac{1}{2})$, and hence $2\pi(\delta - (j-b)/N) \in [-\pi, \pi)$. Using inequality (3.1) then yields

$$|1 - e^{2\pi i(\delta - (j-b)/N)}| \geq \frac{2 \cdot 2\pi|\delta - (j-b)/N|}{\pi} = 4 \left| \delta - \frac{(j-b)}{N} \right| = \frac{4|N\delta - (j-b)|}{N}.$$

Using $|1 - e^{2\pi i(N\delta - (j-b))}| \leq 2$ for the numerator, we then obtain

$$|\alpha_j|^2 \leq \frac{1}{N^2} \frac{4}{(4|N\delta - (j-b)|/N)^2} = \frac{1}{4(N\delta - (j-b))^2}. \quad (3.2)$$

Failure probability. Let $e \in \mathbb{N}$ be a positive integer. In this final part, we will bound the failure probability $\mathbb{P}(|j - b| > e)$.

Since $N\delta \in [0, 1]$, we have for $j - b \geq e + 1$, $(j - b) - N\delta \geq (j - b) - 1$; for $(j - b) \leq -(e + 1)$, $|(j - b) - N\delta| \geq |j - b| - 1$. Applying (3.2) and substituting $m = |j - b| - 1$, we deduce

$$\begin{aligned} \mathbb{P}(|j - b| > e) &\leq \frac{1}{4} \sum_{j \neq b, |j-b| > e} \frac{1}{(N\delta - (j - b))^2} \\ &\leq \frac{1}{4} \left(\sum_{j-b=-2^{t-1}+1}^{-(e+1)} \frac{1}{(|j - b| - 1)^2} + \sum_{j-b=e+1}^{2^{t-1}} \frac{1}{((j - b) - 1)^2} \right) \\ &= \frac{1}{2} \sum_{m=e}^{2^{t-1}-1} \frac{1}{m^2} \leq \frac{1}{2} \int_{e-1}^{\infty} \frac{dm}{m^2} = \frac{1}{2(e-1)}. \end{aligned}$$

Now set $e = 2^{t-n} - 1$, $t \geq n + 1$. Then $|j - b| \leq e = 2^{t-n} - 1$ implies

$$|\tilde{\varphi} - \varphi| = \left| \frac{j - b}{2^t} - \delta \right| \leq \frac{|j - b|}{2^t} + \delta \leq \frac{2^{t-n} - 1}{2^t} + \frac{1}{2^t} = 2^{-n},$$

where we used $\delta < 2^{-t}$. Consequently, with the choice $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$, the failure probability satisfies

$$\mathbb{P}(|j - b| > e) \leq \frac{1}{2(e-1)} = \frac{1}{2(2^{t-n} - 2)} \leq \epsilon,$$

therewith concluding the proof. $\square$

Remark 3.3. The classical DFT approach to frequency estimation requires 2^t sequential evaluations of $s_k = \langle u, U^k u \rangle$, giving circuit depth $O(2^t)$, followed by a classical FFT in $O(t 2^t)$ operations. QPE reduces the depth to $O(t)$: the *dyadic schedule* uses exactly t controlled- U^{2^j} gates — one per clock qubit — to encode all 2^t signal values coherently in the clock amplitudes simultaneously, since phase kickback with the j -th clock qubit accumulates $e^{2\pi i 2^j \varphi}$ in superposition over all 2^t clock states at once. The $\text{QFT}_t^\dagger$ then extracts the dominant frequency φ by interference in $O(t^2)$ additional gates. The total count of U -applications is $O(2^t)$ in both methods; the quantum advantage is the factor- 2^t reduction in *circuit depth* — from $O(2^t)$ sequential matrix-vector products to $O(t)$ parallel rounds of phase kickback — enabled by quantum superposition.

Example 3.4 ($n = 3$). Let $\varphi = 3/8 = [0.011]$. With $t = 3$ clock qubits, $2^t \varphi = 3$ is an integer, so $\epsilon = 0$ and QPE outputs $|3\rangle$ with probability 1. The binary fraction $[0.011]$ encodes φ exactly.

Exercise 3.5. Let $\varphi = 1/3$, $t = 3$. Compute the probability that QPE outputs the nearest 3-bit estimate $k = 3$ (i.e., $\tilde{\varphi} = 3/8$). Then compute the probability of the second-nearest output $k = 2$ ($\tilde{\varphi} = 1/4$) and verify that the total probability exceeds $1/2$.

4 Shor's Factoring Algorithm

Shor's algorithm factors an n -bit integer $N \in \mathbb{N}$ in $O(n^3)$ quantum gate operations, compared to the best known classical algorithm (GNFS), which requires $\exp(O(n^{1/3}(\log n)^{2/3}))$ operations. Shor's algorithm reduces factoring to order-finding, which is solved by QPE. We first show how a period r yields a factor of N (reduction), then develop the quantum order-finding subroutine (QPE applied to $U_{x,N}$), and finally explain how the continued fraction algorithm extracts r from the QPE output.

Reduction from factoring to order-finding

Let $x, N \in \mathbb{N}$ be positive integers with $x < N$ and $\gcd(x, N) = 1$, i.e., x and N are coprime.

Definition 4.1 (Order). The *order* of x modulo N is the least positive integer $r = \text{ord}_N(x)$ such that $x^r \equiv 1 \pmod{N}$.

Proposition 4.2. *Let $N \in \mathbb{N}$ be an odd composite not a prime power. If $r = \text{ord}_N(x)$ is even and $x^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(x^{r/2} \pm 1, N)$ are non-trivial factors of N . A uniformly random x coprime to N satisfies these conditions with probability $\geq 1 - 2^{-(m-1)}$, where m is the number of distinct prime factors of N .*

Theorem 4.3 (Shor's algorithm [1]). *Shor's algorithm factors an n -bit integer N using $O(n^3)$ quantum gates and $O(n^3)$ classical post-processing per run, succeeding with probability $\Omega(1)$ after $O(\log_2 \log_2 N)$ repetitions.*

4.1 Quantum order finding

Order-finding is believed to be classically hard: no algorithm is known to run in time polynomial in $d = \lceil \log_2 N \rceil$, which brings us to the quantum order-finding algorithm.

Algorithm 4.1 Quantum Order Finding

Input: Integers x, N with $\gcd(x, N) = 1$; unitary $U|y\rangle = |xy \bmod N\rangle$;

$t = 2d + 1 + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ clock qubits; $d = \lceil \log_2 N \rceil$ system qubits

Output: Order r such that $x^r \equiv 1 \pmod{N}$

- 1: Initialise $|0\rangle^{\otimes t}|1\rangle$ ▷ By (4.1), $|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle$
 - 2: Apply $H^{\otimes t}$ to clock: $\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle|1\rangle$
 - 3: **for** $k = 0, 1, \dots, t-1$ **do**
 - 4: Apply controlled- U^{2^k} (cf. Remark 3.1)
 - 5: **end for** ▷ System state $\approx \frac{1}{\sqrt{r}} \sum_s (\frac{1}{\sqrt{2^t}} \sum_j e^{2\pi i s j / r} |j\rangle) |u_s\rangle$
 - 6: Apply $\text{QFT}_t^\dagger$ to clock register
 - 7: **Measure** clock; obtain k with $k/2^t \approx s/r$
 - 8: Apply continued fractions algorithm to $k/2^t$; obtain candidate r'
 - 9: **if** $x^{r'} \equiv 1 \pmod{N}$ **then return** r'
 - 10: **else repeat**
 - 11: **end if**
-

The algorithm's procedure is a slight modification of the QPE. As in QPE, there is the clock qubit register and the system register.

The unitary. The system register holds d qubits. We write $|y\rangle$ for the computational basis state indexed by the integer $y \in \{0, \dots, 2^d - 1\}$; this is the tensor product $|y_{d-1}\rangle \otimes \dots \otimes |y_0\rangle$ of the bits of the binary representation $y = [y_{d-1} \dots y_0]_2$. Define the unitary operator U on $\mathbb{C}^{2^d}$ by

$$U|y\rangle = |xy \bmod N\rangle, \quad 0 \leq y \leq N-1,$$

and $U|y\rangle = |y\rangle$ for $N \leq y \leq 2^d - 1$. Here $|xy \bmod N\rangle$ is a computational basis state: the qubit string encoding the integer $xy \bmod N \in \{0, \dots, N-1\}$ in binary. In full:

$$U|y_{d-1} \dots y_0\rangle = |z_{d-1} \dots z_0\rangle, \quad z = xy \bmod N.$$

Since $\gcd(x, N) = 1$, multiplication by x is a bijection on $\{0, \dots, N-1\}$ (every element has a unique inverse $x^{-1} \bmod N$), so U is a permutation of the computational basis, hence unitary. Its inverse is $U^{-1}|y\rangle = |x^{-1}y \bmod N\rangle$.

Eigenstates. Let $r = \text{ord}_N(x)$ be the order of x modulo N . For $0 \leq s \leq r-1$, define

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |x^k \bmod N\rangle.$$

Since $\gcd(x, N) = 1$, each $|x^k \bmod N\rangle$ is a computational basis state.

Proposition 4.4 (Eigenstructure of U). $U|u_s\rangle = e^{2\pi i s / r} |u_s\rangle$ for each $s = 0, \dots, r-1$.

Proof. The proof follows by simply computing

$$\begin{aligned} U|u_s\rangle &= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |x^{k+1} \bmod N\rangle \\ &= e^{2\pi i s / r} \cdot \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s (k+1) / r} |x^{k+1} \bmod N\rangle = e^{2\pi i s / r} |u_s\rangle, \end{aligned}$$

where we used the periodicity $x^r \equiv 1 \pmod{N}$. □

Therefore, each eigenphase is $\varphi_s = s/r$. In particular, the order $r = \text{ord}_N(x)$ appears as the common denominator of the eigenphases. QPE applied to U , and any eigenvector $|u_s\rangle$ then provides an estimate to φ_s . To obtain the denominator r , one then applies continued fractions.

One issue we still have, however, is preparing the eigenvector $|u_s\rangle$, which is unknown because it requires knowing the order r . Fortunately, any eigenvector will do just fine in determining r , in which case, the following trick can help us.

The $|1\rangle$ trick. The key observation is that

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = \sum_{k=0}^{r-1} \left(\frac{1}{r} \sum_{s=0}^{r-1} e^{-2\pi i s k / r} \right) |x^k \bmod N\rangle = |1\rangle, \quad (4.1)$$

where the last equality holds since for $k = 0$,

$$\frac{1}{r} \sum_{s=0}^{r-1} e^{-2\pi i s k / r} = 1 \quad \text{and} \quad \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle \quad \text{is a unit vector.}$$

Preparing the second register in $|1\rangle$ — trivial to construct — is therefore equivalent to preparing a uniform superposition over all eigenstates $|u_s\rangle$.

Measurement. The measurement yields a random $s \in \{0, \dots, r-1\}$ uniformly. The continued fraction algorithm (below) recovers r from $\tilde{\varphi} \approx s/r$, succeeding when $\gcd(s, r) = 1$. By [1, Theorem A4.13], a random x coprime to N gives even r with $x^{r/2} \not\equiv -1$ with probability $\geq 1/2$. The probability that $\gcd(s, r) = 1$ is of the order $\Omega(1/\log \log N)$, so $O(\log \log N)$ repetitions suffice — at most ≈ 4 in practice for a 2048-bit N .

Exercise 4.5. (a) Prove equation (4.1). *Hint:* substitute the definition of $|u_s\rangle$ and use the identity $\sum_{s=0}^{r-1} e^{-2\pi i s k/r} = r \delta_{k,0}$.

(b) Prove the generalisation: for each $k = 0, \dots, r-1$,

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{2\pi i s k/r} |u_s\rangle = |x^k \bmod N\rangle.$$

Interpret this as the inverse QFT of the sequence $(|u_0\rangle, |u_1\rangle, \dots, |u_{r-1}\rangle)$ evaluated at k .

Continued fractions

The continued fractions algorithm represents real numbers using only integers, via expressions of the form

$$[a_0, \dots, a_M] := a_0 + \frac{1}{a_1 + \frac{1}{\dots + \frac{1}{a_M}}}, \quad a_0, \dots, a_M \in \mathbb{N}.$$

The m th-convergent ($0 \leq m \leq M$) to this continued fraction is given by $[a_0, \dots, a_m]$.

It turns out that if $\varphi = s/r$ is a rational number and s, r are d -bit integers, then the continued fraction expansion of φ can be computed using $O(d^3)$ operations. More precisely, the following statement holds:

Theorem 4.6 (Best rational approximation). *If $|s/r - \tilde{\varphi}| \leq 1/(2r^2)$, then s/r is a convergent of the continued fraction expansion of $\tilde{\varphi}$ and can be computed in $O(d^3)$ operations using the continued fractions algorithm.*

After measuring, the quantum order-finding algorithm produces a phase $\tilde{\varphi}$ with $|\tilde{\varphi} - s/r| \leq 2^{-t}$. Since $r \leq N \leq 2^d$ and $t = 2d + O(1)$, we have $|\tilde{\varphi} - s/r| \leq 1/(2r^2)$. Hence, Theorem 4.6 applies, allowing us to compute the order r in $O(d^3)$ operations.

Remark 4.7. The continued fraction step is a classical rational approximation problem: given $\tilde{\varphi}$, find the best fraction p/q with $q \leq N$. The Euclidean algorithm underlies the recurrence, and the condition $t = 2d + O(1)$ is chosen precisely so that Theorem 4.6 applies with $r \leq N$.

Example 4.8 (Taken from [1]). The order of $x = 7 \bmod N = 15$ is $r = 4$ (since $7^4 = 2401 \equiv 1$). Take $t = 11$ clock qubits. The eigenstates have phases $s/r \in \{0, 1/4, 2/4, 3/4\}$. QPE outputs one of $\{0, 512, 1024, 1536\}$ (each with probability $1/4$). Suppose we obtain $k = 1536$; then $\varphi = 1536/2048 = 3/4$, and continued fractions gives the convergent $3/4$, hence $r = 4$. Since r is even and $7^2 \bmod 15 = 4 \not\equiv -1 \bmod 15$, we deduce

$$\gcd(7^2 - 1, 15) = \gcd(48, 15) = 3 \quad \text{and} \quad \gcd(7^2 + 1, 15) = \gcd(50, 15) = 5. \quad \checkmark$$

5 Grover's Algorithm and Amplitude Estimation

Having seen QPE extract an eigenphase (Section 3) and Shor use it to find a hidden period (Section 4), we now study amplitude amplification. Grover's algorithm amplifies the amplitude of a marked subspace by geometric rotation, and amplitude estimation extracts the size of that subspace by applying QPE to the Grover operator. The section closes with a connection to Hamiltonian simulation: the Stone generator $\mathcal{G}$ of the Grover operator $G = e^{i\mathcal{G}}$ anticipates the structure of product formulas in Section 6.

5.1 Grover's algorithm

Grover's algorithm [1] searches an unstructured database of $N = 2^n$ items containing M marked elements, using $O(\sqrt{N/M})$ queries to a quantum oracle.

Motivation. Without any structure on f , a classical algorithm must inspect $\Omega(N/M)$ items on average before finding a marked one. This is tight — random sampling with replacement finds a marked element in $O(N/M)$ queries. Grover's algorithm achieves $O(\sqrt{N/M})$, a quadratic speedup, by exploiting interference: it coherently amplifies the amplitudes of marked states while suppressing those of unmarked states. This is analogous to an iterative method that squares the spectral gap at each step — the convergence is determined by the angle $\theta \approx \sqrt{M/N}$, and $O(1/\theta)$ iterations suffice to reach the target.

Setup. Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a function that marks M elements ($f(x) = 1$ if x is marked and $f(x) = 0$ otherwise). The *phase oracle* O_f acts as $O_f|x\rangle = (-1)^{f(x)}|x\rangle$.

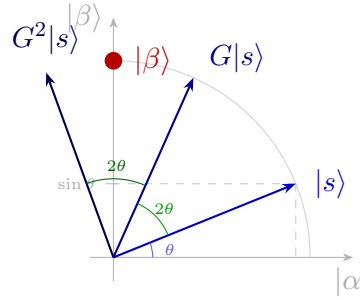


Figure 5.1: Geometric picture of Grover's algorithm in the plane $\text{span}\{|\alpha\rangle, |\beta\rangle\}$. The initial state $|s\rangle$ lies at angle θ from the unmarked axis $|\alpha\rangle$, where $\sin \theta = \sqrt{M/N}$. Each application of G rotates the state by 2θ towards the marked superposition $|\beta\rangle$. After $k = \lfloor \pi/(4\theta) \rfloor$ steps the state is close to $|\beta\rangle$, and measurement yields a marked element with high probability.

Geometric picture. See Figure 5.1. Define the two orthogonal unit vectors

$$|\alpha\rangle = \frac{1}{\sqrt{N-M}} \sum_{f(x)=0} |x\rangle \quad (\text{unmarked superposition}), \quad |\beta\rangle = \frac{1}{\sqrt{M}} \sum_{f(x)=1} |x\rangle \quad (\text{marked}).$$

Then, the initial state $|s\rangle = H^{\otimes n}|0^n\rangle$ satisfies:

$$|s\rangle = \sqrt{(N-M)/N} |\alpha\rangle + \sqrt{M/N} |\beta\rangle,$$

which lies in the plane $V_{\alpha,\beta} := \text{span}\{|\alpha\rangle, |\beta\rangle\}$ at angle θ from $|\alpha\rangle$, where $\sin \theta = \sqrt{M/N}$.

The so-called *Grover operator* $G = D \cdot O_f$ is the operator consisting of:

- O_f : reflection about $|\alpha\rangle$ (flips sign of $|\beta\rangle$ component);
- $D = 2|s\rangle\langle s| - \mathbb{I}$: reflection about $|s\rangle$.

Two reflections compose a rotation by 2θ in the $V_{\alpha,\beta}$ -plane.

Implementing the diffusion operator. The oracle O_f is supplied as a black box, but the reflection $D = 2|s\rangle\langle s| - \mathbb{I}$ about $|s\rangle$ must be built from elementary gates. Since $|s\rangle = H^{\otimes n}|0^n\rangle$ and $H^{\otimes n}$ is its own inverse,

$$D = H^{\otimes n}(2|0^n\rangle\langle 0^n| - \mathbb{I})H^{\otimes n}.$$

The inner operator fixes $|0^n\rangle$ and flips the sign of every other computational-basis state. It is, up to a global sign, the phase flip of $|0^n\rangle$ alone, realised by sandwiching an $(n-1)$ -controlled- Z between layers of X :

$$2|0^n\rangle\langle 0^n| - \mathbb{I} = -X^{\otimes n}C^{n-1}ZX^{\otimes n},$$

where $C^{n-1}Z$ flips the sign of $|1^n\rangle$ and the X layers map $|0^n\rangle \leftrightarrow |1^n\rangle$. Reading right to left, D is therefore the fixed, oracle-independent circuit $H^{\otimes n}, X^{\otimes n}, C^{n-1}Z, X^{\otimes n}, H^{\otimes n}$, using $2n$ Hadamards, $2n$ X gates, and one $(n-1)$ -controlled- Z (itself $O(n)$ Toffoli gates with ancillas). The leading global phase -1 replaces D by $-D$, which affects neither the measurement probabilities nor the 2θ rotation of Figure 5.1.

Measurement. After $k = \lfloor \pi/(4\theta) \rfloor$ applications of G , the state is close to $|\beta\rangle$ and one measurement of the n -qubit register yields a marked element with probability $\geq 1 - M/N$. The measurement costs $O(n)$ classical bits and is performed exactly once per run; no repetition is needed when M/N is bounded away from zero.

If M is unknown — the typical case — the optimal iteration count $k = \lfloor \pi/(4\theta) \rfloor$ is unknown as well; the discussion below handles this without knowing M in advance.

Algorithm 5.1 Grover Search

Input: Phase oracle O_f for $f : \{0, 1\}^n \rightarrow \{0, 1\}$; number of marked elements M ; $N = 2^n$

Output: A marked element x with $f(x) = 1$, with probability $\geq 1 - M/N$

- 1: Initialise $|\psi\rangle = H^{\otimes n}|0^n\rangle$
 - 2: Set $k = \lfloor \pi/(4\theta) \rfloor$ where $\theta = \arcsin \sqrt{M/N}$
 - 3: **for** $j = 1, 2, \dots, k$ **do**
 - 4: Apply oracle: $|\psi\rangle \leftarrow O_f|\psi\rangle$
 - 5: Apply reflection: $|\psi\rangle \leftarrow (2|s\rangle\langle s| - \mathbb{I})|\psi\rangle$
 - 6: **end for**
 - 7: **Measure** in computational basis; **return** result
-

Theorem 5.1 (Grover's algorithm). *After $k = \lfloor \pi/(4\theta) \rfloor$ applications of G , the state has squared overlap $\geq 1 - M/N$ with $|\beta\rangle$. Measuring then yields a marked element with probability $\geq 1 - M/N$. The number of oracle queries is $O(\sqrt{N/M})$.*

Proof. In the $\{|\alpha\rangle, |\beta\rangle\}$ basis, the initial state is at angle θ from $|\alpha\rangle$, and each application of G rotates the state by 2θ . After k steps the angle from $|\alpha\rangle$ is $(2k+1)\theta$. Therefore, choosing $k = \lfloor \pi/(4\theta) \rfloor$ makes $(2k+1)\theta \approx \pi/2$, so the state is close to $|\beta\rangle$. Since $\sin \theta = \sqrt{M/N}$, we have $\theta \geq \sqrt{M/N}$ for small θ , giving $k = O(1/\theta) = O(\sqrt{N/M})$. $\square$

Remark 5.2 (Lower bound [2]). The $O(\sqrt{N/M})$ query complexity is optimal: any quantum algorithm requires $\Omega(\sqrt{N/M})$ oracle queries to find a marked element with constant probability.

Unknown number of solutions. Algorithm 5.1 needs θ , hence M , to fix the iteration count. When M is unknown one cannot simply “iterate as long as possible”: after k steps the success probability

$$p_k = \sin^2((2k+1)\theta)$$

oscillates in k . Too few iterations leave the marked amplitude small; too many *overshoot* $|\beta\rangle$, rotating past it so that p_k falls again. The standard remedy is to determine θ before searching.

Quantum counting (Exercise 5.3) applies QPE to the Grover operator G , whose eigenphase on $V_{\alpha,\beta}$ is 2θ , to estimate θ — hence $M = N \sin^2 \theta$ — and then runs Algorithm 5.1 once with $k = \lfloor \pi/(4\theta) \rfloor$. Estimating θ to enough bits to pin down k costs $O(\sqrt{N})$ applications of G .

Exercise 5.3 (Quantum counting). (a) Show that the Grover operator G has eigenvalues $e^{\pm 2i\theta}$ with eigenvectors

$$|e_+\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle - i|\beta\rangle) \quad \text{and} \quad |e_-\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle + i|\beta\rangle).$$

In particular, show that G restricted to $V_{\alpha,\beta}$ is a rotation matrix $R_{2\theta}$.

(b) Use this to construct an algorithm to determine the number M of marked elements.

5.2 Amplitude estimation

From quantum counting to amplitude estimation. Quantum counting (Section 5.1) applied QPE to the Grover operator G to read off its eigenphase 2θ , recovering $\sin^2 \theta = M/N$ — the probability that measuring $|s\rangle = H^{\otimes n}|0^n\rangle$ yields a marked element. Nothing in that argument used the uniform state preparation.

In general, one can replace $H^{\otimes n}$ by an arbitrary *state-preparation circuit* A with

$$A|0\rangle = \sqrt{p}|\psi_1\rangle|1\rangle + \sqrt{1-p}|\psi_0\rangle|0\rangle,$$

where the last qubit flags “good” ($|1\rangle$) versus “bad” ($|0\rangle$) outcomes and $p \in [0, 1]$ is the probability to be estimated. Forming the Grover operator

$$G_A = (2|s_A\rangle\langle s_A| - \mathbb{I})O, \quad |s_A\rangle = A|0\rangle,$$

exactly as in Section 5.1 — with oracle O flipping the sign of the $|1\rangle$ component — makes G_A a rotation by 2θ with $\sin^2 \theta = p$ in the plane $\text{span}\{|\psi_0\rangle|0\rangle, |\psi_1\rangle|1\rangle\}$. Running QPE on G_A therefore reads off p . This is *amplitude estimation*, where quantum counting is a special case with $A = H^{\otimes n}$, $p = M/N$.

Measurement. Amplitude estimation ends with a single measurement of the clock register, yielding an integer k from which $\tilde{p} = \sin^2(\pi k/2^t)$ is computed classically. The success probability is at least $8/\pi^2 \approx 0.81$ (Theorem 5.4 below), so $O(1)$ repetitions suffice in expectation. The total number of applications of A is $O(2^t) = O(1/\epsilon)$.

Theorem 5.4 (Amplitude estimation [3]). *QPE applied to G_A with t clock qubits produces an estimate $\tilde{p}$ satisfying*

$$|\tilde{p} - p| \leq \frac{2\pi\sqrt{p(1-p)}}{2^t} + \frac{\pi^2}{2^{2t}},$$

with probability at least $8/\pi^2$, using 2^t applications of A . In particular, to achieve precision ϵ , it suffices to take $t = O(1/\epsilon)$ applications.

Proof sketch. The Grover operator G_A has eigenvalues $e^{\pm 2i\theta}$ with $\sin^2 \theta = p$. QPE estimates $\theta/\pi \in [0, 1]$ to t bits, from which $p = \sin^2(\pi\tilde{\theta})$ is computed. The error bound follows from the QPE precision (Theorem 3.2) and the Lipschitz constant of $\sin^2$. $\square$

6 Hamiltonian Simulation

6.1 Stone's theorem and unitary generators

Every quantum gate is a unitary operator. The question of where unitaries come from is answered by Stone's theorem, which establishes a one-to-one correspondence between strongly continuous one-parameter unitary groups and self-adjoint generators.

Theorem 6.1 (Stone's theorem). *Let $\mathcal{H}$ be a finite dimensional Hilbert space, and let $H \in \mathcal{O}(\mathcal{H})$ be Hermitian ($H^\dagger = H$). Then the family $(U_t)_{t \in \mathbb{R}}$ defined by*

$$U_t := e^{iHt} = \sum_{k=0}^{\infty} \frac{(iHt)^k}{k!}$$

is a one-parameter group of unitary operators, i.e., $U_t^\dagger U_t = \mathbb{I}$, $U_0 = \mathbb{I}$, and $U_s U_t = U_{s+t}$.

Conversely, every strongly continuous one-parameter unitary group $(U_t)_{t \in \mathbb{R}}$ is of this form for a unique Hermitian H , called the Stone generator (or Hamiltonian) of the group.

Remark 6.2. Stone's theorem explains why Hamiltonians in quantum mechanics are Hermitian: unitarity of time evolution requires the generator to be self-adjoint. Conversely, every Hermitian matrix generates a valid unitary evolution. The eigendecomposition $H = V \Lambda V^\dagger$ gives $e^{iHt} = V \text{diag}(e^{i\lambda_1 t}, \dots, e^{i\lambda_N t}) V^\dagger$, so the eigenvalues of H are the frequencies of the corresponding unitary oscillations.

Example 6.3 (Grover operator revisited). Recall the Grover operator G from Section 5. Since G is unitary, Stone's theorem provides a self-adjoint operator $\mathcal{G}$ such that $G = e^{i\mathcal{G}}$. In the plane $V_{\alpha,\beta} := \text{span}\{|\alpha\rangle, |\beta\rangle\}$, the operator G acts as a rotation by 2θ , so in the basis $\{|\alpha\rangle, |\beta\rangle\}$:

$$G|_{V_{\alpha,\beta}} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} = \exp\left(2\theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right) = e^{-i \cdot 2\theta \sigma_y},$$

where σ_y is the Pauli- Y matrix in this basis. The Stone generator restricted to $V_{\alpha,\beta}$ is therefore

$$\mathcal{G}|_{V_{\alpha,\beta}} = -2\theta \sigma_y = 2\theta i (|\alpha\rangle\langle\beta| - |\beta\rangle\langle\alpha|).$$

Hamiltonian simulation with $H = \mathcal{G}$ and integer time $t = k$ gives $G^k = e^{ik\mathcal{G}}$, which is exactly the k -th iterate of Grover's algorithm.

Exercise 6.4. Show that any unitary U with eigenvalues ± 1 takes the form $U = \exp(i\frac{\pi}{2}(\mathbb{I} - U))$, i.e., it has the generator $H = \frac{\pi}{2}(\mathbb{I} - U)$.

Remark 6.5 (Stone generators of the QPE circuit). Every gate in the QPE circuit has an explicit Stone generator. We list them in the $\{|0\rangle, |1\rangle\}$ basis (or its tensor powers).

- *Hadamard gate.* Since $H^2 = \mathbb{I}$, the eigenvalues of H are ± 1 . The Stone generator is $\mathcal{G}_H = \frac{\pi}{2}(\mathbb{I} - H)$. The $+1$ eigenspace of H contributes phase $e^0 = 1$; the -1 eigenspace contributes $e^{i\pi} = -1$. ✓
- *Phase rotation $R_\varphi = \text{diag}(1, e^{i\varphi})$.* This acts as the identity on $|0\rangle$ and as $e^{i\varphi}$ on $|1\rangle$, so the generator is $\mathcal{G}_{R_\varphi} = \varphi|1\rangle\langle 1| = \frac{\varphi}{2}(\mathbb{I} - Z)$. For the controlled- R_m gate, which applies $R_{2\pi/2^m}$ to the target iff the control is $|1\rangle$, the two-qubit generator is

$$\mathcal{G}_{\text{c-}R_m} = \frac{2\pi}{2^m}|1\rangle\langle 1| \otimes |1\rangle\langle 1| = \frac{\pi}{2^{m-1}}|11\rangle\langle 11|.$$

- *Controlled- U^{2^j} gate.* This is the most important case for QPE. If $U = e^{i\mathcal{G}}$ for some Hermitian $\mathcal{G}$, then $U^{2^j} = e^{i2^j\mathcal{G}}$: the generator is $\mathcal{G}$ scaled by 2^j . The controlled version has a two-register generator

$$\mathcal{G}_{c-U^{2^j}} = |1\rangle\langle 1| \otimes 2^j\mathcal{G},$$

acting non-trivially only when the clock qubit is $|1\rangle$. This dyadic scaling $2^j\mathcal{G}$ is the generator-level explanation of QPE: the j -th clock qubit accumulates phase $e^{2\pi i 2^j \varphi}$ because its generator has eigenvalue $2^j\varphi$ (where φ is the eigenvalue of $\mathcal{G}$). The binary digits of φ are thus encoded in the generators of successive controlled- U^{2^j} gates.

- *SWAP gate.* The swap gate $\text{SWAP} = \frac{1}{2}(\mathbb{I} + X \otimes X + Y \otimes Y + Z \otimes Z)$ has eigenvalues ± 1 , giving generator $\mathcal{G}_{\text{SWAP}} = \frac{\pi}{2}(\mathbb{I} - \text{SWAP}) = \frac{\pi}{4}(\mathbb{I} - X \otimes X - Y \otimes Y - Z \otimes Z)$.

Unfortunately, the generator of $\text{QFT}_t^\dagger$ as a whole is not a simple closed form, since the constituent gates do not commute. Instead, $\text{QFT}_t^\dagger$ is understood as a product $V_1 \cdot V_2 \cdots V_K$ of elementary unitaries, each with its own generator as listed above.

The challenge. Given a Hermitian $H \in \mathcal{O}(\mathcal{H})$ and a time t , how efficiently can we implement the unitary e^{iHt} as a quantum circuit? This is also the gateway to HHL (Section 7), where QPE is applied to e^{iAt} for a matrix A to be inverted.

Remark 6.6. The problem of computing $e^{Ht}v$ for a large sparse matrix H and a vector v arises throughout numerical analysis: in the solution of stiff ODEs ($\dot{u} = Hu$, $u(0) = v$), in the computation of matrix functions, and in spectral methods for PDEs. Classically, direct computation of e^H costs $O(N^3)$ via diagonalisation, and approximation via Krylov–Arnoldi methods costs $O(Nk^2)$ for k Arnoldi steps (with $k \ll N$ for sparse H).

In the quantum setting, e^{iHt} must be decomposed into elementary quantum gates (H, CNOT, R_θ) that a quantum computer can execute. The challenge is that a general $N \times N$ unitary has $O(N^2)$ parameters, but a sparse quantum circuit can only implement $O(n \cdot \text{poly}(n))$ gates for $N = 2^n$. Product formulas (Trotter splitting) are the quantum counterpart of operator-splitting methods in numerical ODEs, and they achieve this by exploiting the sparse structure of H .

6.2 Product formulas

The strategy is to decompose $H = \sum_{j=1}^L H_j$ into terms H_j that are individually easy to exponentiate, and approximate e^{iHt} by a product of simple exponentials.

Theorem 6.7 (First-order Trotter formula). *For any Hermitian matrices A, B and $t \in \mathbb{R}$,*

$$e^{i(A+B)t} = \lim_{r \rightarrow \infty} (e^{iAt/r} e^{iBt/r})^r.$$

For finite r , the error is

$$\|e^{i(A+B)t} - (e^{iAt/r} e^{iBt/r})^r\| \leq \frac{t^2 \|[A, B]\|}{2r},$$

where $\|\cdot\|$ denotes the operator norm.

Remark 6.8. The first-order Trotter formula is the quantum analog of the Lie–Trotter operator splitting in numerical ODEs. The error bound via the Baker–Campbell–Hausdorff formula (BCH) mirrors the local truncation error analysis of splitting schemes: $\|[A, B]\|$ measures the failure of commutativity, just as the commutator term appears in the modified equation of Strang splitting.

Theorem 6.9 (Strang (second-order) splitting). *The symmetric product formula*

$$S_2(t) = e^{iAt/2} e^{iBt} e^{iAt/2}$$

satisfies

$$\|e^{i(A+B)t} - S_2^r(t/r)\| \leq \frac{t^3(\|[A, [A, B]]\| + \|[B, [A, B]]\|)}{24r^2},$$

a second-order error in $1/r$.

6.3 Sparse Hamiltonians

An $N \times N$ Hermitian matrix ($N = 2^n$) is d -sparse if each row has at most d of its N entries non-zero; the regime of interest is $d \ll N$, typically $d = O(\text{poly}(n))$. In practice H is accessed not as an explicit matrix but through two reversible *sparse oracles*: a *sparsity oracle* $O_\nu|j\rangle|\ell\rangle = |j\rangle|\nu(j, \ell)\rangle$ returning the column of the ℓ -th non-zero entry of row j , and a *value oracle* $O_H|j\rangle|k\rangle|0\rangle = |j\rangle|k\rangle|H_{jk}\rangle$ returning the entry itself. A *query* is one use of either, and simulation cost is counted in queries.

Sparse simulation reduces H to easy pieces. A locally computable edge colouring of the graph of non-zero entries writes $H = \sum_{c=1}^m H_c$ with $m = O(d^2)$, where each H_c is 1 -sparse (at most one non-zero per row and column) and accessible in $O(1)$ queries. A 1 -sparse H_c is block diagonal with 2×2 blocks coupling pairs $|j\rangle, |k\rangle$, so $e^{iH_c t}$ is a single controlled rotation at $O(1)$ queries for any t . Applying the product formulas of Section 6.2 to $H = \sum_c H_c$ then gives the following.

Theorem 6.10 (Sparse Hamiltonian simulation). *Let H be a d -sparse Hermitian matrix with $\|H\|_{\max} \leq 1$. Then e^{iHt} can be implemented to precision ε using $O\left(\frac{d^2\|H\|_{\max}t}{\varepsilon^{1/2k}}\right)$ oracle queries, for any $k \geq 1$.*

Remark 6.11. More sophisticated methods (LCU, qubitisation [3]) achieve near-optimal query complexity $O(d\|H\|_{\max}t + \log(1/\varepsilon))$, but Trotter-based methods suffice for the present course.

Example 6.12 (Finite-difference Laplacian). Discretising $-\partial_x^2$ on a periodic grid of $N = 2^n$ points (spacing $h = 1$) gives the matrix L with $L_{jj} = 2$ and $L_{j, j\pm 1} = -1$ (indices mod N). Every row has only three non-zero entries, so L is 3 -sparse independently of N — a D -dimensional grid gives $d = 2D + 1$ — and the oracles are mere index arithmetic, since the non-zeros of row j sit at columns j and $j \pm 1 \bmod N$. Splitting the nearest-neighbor bonds by parity,

$$H_{\text{even}} = \sum_{m=0}^{N/2-1} (|2m\rangle\langle 2m+1| + |2m+1\rangle\langle 2m|), \quad H_{\text{odd}} = \sum_{m=0}^{N/2-1} (|2m+1\rangle\langle 2m+2| + |2m+2\rangle\langle 2m+1|)$$

(indices mod N) yields a 1 -sparse decomposition that is explicit so that $L = 2\mathbb{I} - (H_{\text{even}} + H_{\text{odd}})$. The split is what makes the pieces easy: the full hopping is not block diagonal, since each site lies in *two* bonds, so a single matching cannot cover the chain. Each parity class, by contrast, is a *perfect matching*, i.e., every site in exactly one bond, so H_{even} and H_{odd} are direct sums of the 2×2 exchange block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ over their bonded pairs, hence 1 -sparse and involutory. Exponentiating therefore acts independently on each bond $\{a, b\}$,

$$e^{iH_\bullet t}|_{\text{span}\{|a\rangle, |b\rangle\}} = \begin{pmatrix} \cos t & i \sin t \\ i \sin t & \cos t \end{pmatrix},$$

a single two-level rotation.

The diagonal $2\mathbb{I}$ is a global phase, and only $H_{\text{even}}, H_{\text{odd}}$ fail to commute, so

$$e^{-iLt} = e^{-2it} e^{i(H_{\text{even}}+H_{\text{odd}})t} \approx e^{-2it} (e^{iH_{\text{even}}t/r} e^{iH_{\text{odd}}t/r})^r$$

is exactly the even–odd *split-step* scheme of numerical analysis. It simulates e^{-iLt} — Schrödinger evolution $i\partial_t\psi = -\partial_x^2\psi$, or the e^{iAt} inside HHL for the Poisson equation $-\partial_x^2 u = f$ — in $\text{poly}(n)$ gates, versus the 2^{2n} entries a dense treatment would touch.

Exercise 6.13. Using the first-order Trotter formula, show that e^{iHt} can be simulated to precision ε using $r = \lceil t^2 \max_{j<l} \|[H_j, H_l]\|/\varepsilon \rceil$ Trotter steps, each requiring L exponentials.

Exercise 6.14 (★). Prove Theorem 6.9 by expanding $S_2(t)$ via BCH and showing that the leading error term is of order t^3 .

7 The HHL Algorithm

Solving $Ax = b$ is the central problem of numerical linear algebra. For dense $N \times N$ systems, Gaussian elimination costs $O(N^3)$. For sparse systems, iterative methods such as conjugate gradient (CG) cost $O(N\sqrt{\kappa} \log(1/\epsilon))$ per solve (for symmetric positive definite A with condition number κ), or $O(N\kappa \log(1/\epsilon))$ for general methods such as GMRES. For very large N (e.g. $N = 10^9$ arising from discretization of a 3D PDE), even $O(N)$ operations per iteration are expensive, motivating interest in sublinear methods.

The *HHL algorithm* [4] solves the quantum linear systems problem: given a Hermitian positive definite matrix A and a quantum state $|b\rangle$, prepare $|x\rangle \propto A^{-1}|b\rangle$. HHL's complexity is $O(\log(N) \cdot \kappa^2/\epsilon)$, which is *exponentially* better in N than classical methods (the $1/\epsilon$ can be improved to $\log(1/\epsilon)$; see Remark 7.6). The price paid is twofold: the output is a *quantum state* $|x\rangle$ rather than the classical vector x , and the algorithm requires efficient quantum access to A and b . Understanding these caveats is as important as the speedup itself, and we treat them carefully in Section 7.2.

Key idea. The solution $x = A^{-1}b$ can be written as $x = \sum_j \beta_j \lambda_j^{-1} u_j$, where $Au_j = \lambda_j u_j$ is the spectral decomposition of A and $b = \sum_j \beta_j u_j$. If we could encode the eigenvalues λ_j in a quantum register and then apply the map $\lambda_j \mapsto 1/\lambda_j$, we would obtain $|x\rangle$. This is precisely what HHL does: QPE encodes λ_j , a controlled rotation implements the inversion, and inverse QPE uncomputes the eigenvalue register. The inversion step $\lambda \mapsto 1/\lambda$ is the quantum counterpart of the spectral representation of the matrix inverse, and the condition number κ controls both the gate count (via QPE precision) and the success probability.

Definition 7.1 (Quantum linear systems problem). Given:

- (1) a Hermitian positive definite $A \in \mathbb{C}^{N \times N}$, $N = 2^n$, with eigenvalues $\lambda_j \in [\lambda_{\min}, \lambda_{\max}]$ and condition number $\kappa = \lambda_{\max}/\lambda_{\min}$;
- (2) a quantum state $|b\rangle = \sum_j \beta_j |u_j\rangle$ (in the eigenbasis $\{u_j\}$ of A).

The *quantum linear systems problem* asks to prepare $|x\rangle \propto A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$.

7.1 The algorithm

The HHL algorithm uses three registers: a clock register $\mathcal{C}$ of t qubits to represent the eigenvalues, the system register $\mathcal{S}$ holding $|b\rangle$, and a single ancilla qubit $\mathcal{A}$.

Step 1: QPE on e^{iAt_0} . Since $e^{iAt_0}|u_j\rangle = e^{i\lambda_j t_0}|u_j\rangle$, QPE encodes the eigenvalue λ_j in the clock register:

$$\text{QPE} : |0^t\rangle|b\rangle|0\rangle \mapsto \sum_j \beta_j |\tilde{\lambda}_j\rangle |u_j\rangle |0\rangle,$$

where $\tilde{\lambda}_j$ is a t -bit approximation to $\lambda_j t_0 / 2\pi$, and t_0 is chosen so that $\lambda_j t_0 / (2\pi) \in [0, 1)$. Here $|\tilde{\lambda}_j\rangle$ is shorthand for the computational-basis state $|2^t \tilde{\lambda}_j\rangle$ indexed by the integer $2^t \tilde{\lambda}_j \in \{0, \dots, 2^t - 1\}$, i.e., the same readout convention as in QPE (Section 3), where the clock returns $|2^t \tilde{\varphi}\rangle$. The estimate $\tilde{\lambda}_j$ is then recovered by dividing the stored integer by 2^t .

Step 2: Controlled rotation. The clock now controls the ancilla's rotation. Since the clock is in the superposition $\sum_j \beta_j |\tilde{\lambda}_j\rangle |u_j\rangle$, this is a *uniformly-controlled rotation*

$$\sum_{\ell=0}^{2^t-1} |\ell\rangle \langle \ell|_{\mathcal{C}} \otimes R_y(\theta_\ell)_{\mathcal{A}}, \quad \sin \frac{\theta_\ell}{2} = \frac{C}{\tilde{\lambda}_\ell},$$

Algorithm 7.1 HHL Quantum Linear Solver

Input: Hermitian PD matrix A (sparse oracle); state $|b\rangle$; condition number κ ; precision ϵ

Output: State $|x\rangle \propto A^{-1}|b\rangle$ to precision ϵ

- 1: Initialise clock $\mathcal{C}$, system $\mathcal{S} = |b\rangle$, ancilla $\mathcal{A} = |0\rangle$, $t_0 = 2\pi/\lambda_{\max}$
 - 2: **QPE:** apply QPE with $U = e^{iAt_0}$, via Hamiltonian simulation (Section 6)
$$\triangleright \sum_j \beta_j |u_j\rangle \rightarrow \sum_j \beta_j |\tilde{\lambda}_j\rangle |0^t\rangle |u_j\rangle$$
 - 3: **Controlled rotation:** conditioned on $\tilde{\lambda}_j$, rotate ancilla $|0\rangle \rightarrow \sqrt{1 - C^2/\tilde{\lambda}_j^2} |0\rangle + (C/\tilde{\lambda}_j) |1\rangle$
 - 4: **Inverse QPE:** apply $\text{QPE}^\dagger$ to uncompute clock $\mathcal{C}$
 - 5: **Measure** ancilla $\mathcal{A}$
 - 6: **if** outcome = 1 **then**
 - 7: **return** system register $|x\rangle \propto \sum_j (\beta_j/\lambda_j) |u_j\rangle = A^{-1}|b\rangle$
 - 8: **else**
 - 9: Repeat from Step 1
 - 10: **end if**
-

with the clock as control, where the value ℓ that is read off the clock selects the angle θ_ℓ applied to the ancilla, and the unitary acts on each ℓ . On the branch where the clock holds $|\tilde{\lambda}_j\rangle$ (so $\ell = 2^t \tilde{\lambda}_j$), the ancilla is rotated:

$$|\tilde{\lambda}_j\rangle |u_j\rangle |0\rangle \mapsto |\tilde{\lambda}_j\rangle |u_j\rangle \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right),$$

where $C \leq \lambda_{\min}$ is a normalization constant. Therefore, this maps $\lambda_j \mapsto C/\lambda_j$ in the $|1\rangle$ amplitude, effecting the inversion A^{-1} .

Remark 7.2 (Implementing controlled rotations). The required rotation is $\sin(\theta_j/2) = C/\tilde{\lambda}_j$, so the angle θ_j depends non-linearly on the integer stored in the clock register. Nevertheless, it can be implemented in three sub-steps:

- (2a) *Compute the reciprocal.* Using quantum arithmetic (reversible integer division), compute $\mu_j := \widetilde{C/\lambda_j}$ to s bits of fixed-point precision in a fresh ancilla register:

$$|\tilde{\lambda}_j\rangle |0^s\rangle \mapsto |\tilde{\lambda}_j\rangle |\mu_j\rangle.$$

- (2b) *Rotate using the reciprocal.* Write $\mu_j = \sum_{k=0}^{s-1} c_k 2^{-k}$. The rotation is parametrized directly by its matrix entries $\sin(\theta_j/2) = \mu_j$ and $\cos(\theta_j/2) = \sqrt{1 - \mu_j^2}$, both algebraic functions of μ_j . Apply controlled- R_y gates conditioned on the bits c_k :

$$\prod_{k=0}^{s-1} \text{c-}R_y(c_k 2^{-(k-1)}) |0\rangle \approx \sqrt{1 - \mu_j^2} |0\rangle + \mu_j |1\rangle.$$

- (2c) *Uncompute.* Apply the inverse of Step 2a to return the reciprocal register to $|0^s\rangle$, disentangling it before inverse QPE.

Step 3: Inverse QPE. Revert the clock register by applying $\text{QPE}^\dagger$, returning the clock to $|0^t\rangle$:

$$\sum_j \beta_j |0^t\rangle |u_j\rangle \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right).$$

Step 4: Post-selection. Finally, measure the ancilla. In case of the outcome $|1\rangle$ (success), the system register collapses to

$$|x\rangle \propto \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle = A^{-1}|b\rangle.$$

Theorem 7.3 (HHL algorithm). *Let A be a Hermitian positive definite $N \times N$ matrix with condition number κ and $\|A\| \leq 1$. Given quantum access to $|b\rangle$ and a sparse oracle for A , the HHL algorithm prepares $|x\rangle \propto A^{-1}|b\rangle$ to precision ϵ in time $O(\log(N) \kappa^2/\epsilon)$. The success probability of post-selection is $\Theta(C^2/\kappa^2 \lambda_{\min}^2)$.*

Lemma 7.4 (QPE step). *With $t = O(\log(\kappa/\epsilon))$ clock qubits, QPE produces $|\tilde{\lambda}_j - \lambda_j| \leq \epsilon \lambda_{\min}$ for each j , at a cost of $O(2^t) = O(\kappa/\epsilon)$ applications of e^{iAt_0} .*

Lemma 7.5 (Inversion step). *The controlled rotation $\lambda_j \mapsto C/\lambda_j$ introduces an additional error $O(\epsilon)$ in the output state when $|\tilde{\lambda}_j - \lambda_j| \leq \epsilon \lambda_{\min}$.*

Proof of Theorem 7.3. Correctness. Take $C = \lambda_{\min}$ and suppose first that phase estimation is exact ($\tilde{\lambda}_j = \lambda_j$). After Steps 1–3 the clock has been uncomputed to $|0^t\rangle$, leaving the joint system-ancilla state

$$\sum_j \beta_j |u_j\rangle \left(\sqrt{1 - C^2/\lambda_j^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right).$$

Post-selecting the ancilla on $|1\rangle$ keeps the second branch and renormalises, leaving the system register in

$$\frac{\sum_j (C\beta_j/\lambda_j) |u_j\rangle}{\|\sum_j (C\beta_j/\lambda_j) |u_j\rangle\|} = \frac{\sum_j (\beta_j/\lambda_j) |u_j\rangle}{\|\sum_j (\beta_j/\lambda_j) |u_j\rangle\|} \propto A^{-1}|b\rangle,$$

the constant C cancelling in the normalisation — so the ideal output is exactly $|x\rangle$.

Success probability. The outcome $|1\rangle$ occurs with probability equal to the squared norm of that branch,

$$p_{\text{succ}} = \sum_j |\beta_j|^2 \frac{C^2}{\lambda_j^2} = C^2 \|A^{-1}|b\rangle\|^2.$$

Since $\sum_j |\beta_j|^2 = 1$ and $\lambda_j \in [\lambda_{\min}, \lambda_{\max}]$, this lies in $[C^2/\lambda_{\max}^2, C^2/\lambda_{\min}^2]$; the worst case ($|b\rangle$ aligned with the $\lambda_{\max}$ eigenvector) gives $p_{\text{succ}} = \Theta(C^2/\lambda_{\max}^2) = \Theta(C^2/\kappa^2 \lambda_{\min}^2)$, which for $C = \lambda_{\min}$ is $\Theta(1/\kappa^2)$. Repeat-until-success therefore needs $O(\kappa^2)$ runs, reduced to $O(\kappa)$ by amplitude amplification.

Precision and cost. For the inversion to be accurate to relative error ϵ the estimate must obey $|\tilde{\lambda}_j - \lambda_j| \leq \epsilon \lambda_{\min}$ (Inversion lemma), since $\frac{d}{d\lambda} \lambda^{-1} = -\lambda^{-2}$ is largest in magnitude at $\lambda_{\min}$. In phase units $\varphi_j = \lambda_j/\lambda_{\max} \in [1/\kappa, 1]$ this is an additive error ϵ/κ , so by the QPE precision bound (Theorem 3.2) it requires $t = O(\log(\kappa/\epsilon))$ clock qubits and hence $O(2^t) = O(\kappa/\epsilon)$ applications of e^{iAt_0} per run (QPE lemma), each a sparse Hamiltonian simulation of cost $\tilde{O}(\log N)$ (Section 6). Multiplying the per-run cost by the $O(\kappa)$ amplitude-amplification rounds gives $O(\log(N) \kappa^2/\epsilon)$. Finally, the residual QPE error perturbs each rotation angle, and hence the output state, by $O(\epsilon)$ in norm (Inversion lemma), so the prepared state is ϵ -close to $|x\rangle$. $\square$

Measurement. HHL involves two distinct measurement events with very different costs.

Post-selection on the ancilla. After inverse QPE, the single ancilla qubit is measured. The algorithm succeeds only on outcome $|1\rangle$, which occurs with probability $\Theta(C^2/\kappa^2 \lambda_{\min}^2)$ (Theorem 7.3). With the canonical choice $C = \lambda_{\min}$, this is $\Theta(1/\kappa^2)$. To obtain one successful run, the entire circuit must be repeated $O(\kappa^2)$ times — this is a genuine cost and appears explicitly in

the overall complexity. It can be reduced to $O(\kappa)$ by more sophisticated amplitude amplification techniques, but not eliminated.

Estimating a functional of $|x\rangle$. Once $|x\rangle \propto A^{-1}|b\rangle$ has been prepared, any linear functional $\langle x|M|x\rangle$ can be estimated to precision ϵ using $O(1/\epsilon^2)$ measurements (standard sampling) or $O(1/\epsilon)$ measurements (amplitude estimation applied to $|x\rangle$). The classical vector x itself cannot be recovered without $O(N)$ measurements (full tomography), which destroys the exponential speedup in N . HHL therefore provides a quantum advantage only when a small number of scalar functionals of x are required.

Remark 7.6 (Beyond HHL: modern developments). The QPE-based algorithm above is the original and most transparent version; it has since been improved on essentially every axis. The headline developments:

- *Condition number $\kappa^2 \rightarrow \kappa$.* Variable-time amplitude amplification (Ambainis 2012) makes the dependence nearly linear, matching the $\Omega(\kappa)$ lower bound.
- *Precision $1/\epsilon \rightarrow \log(1/\epsilon)$.* Approximating A^{-1} by a linear combination of Hamiltonian-simulation steps (LCU; Childs–Kothari–Somma 2017) removes the lossy phase-estimation step and improves the error dependence exponentially.
- *No phase estimation at all.* Quantum singular value transformation with qubitisation (Low–Chuang 2017; Gilyén–Su–Low–Wiebe 2019) applies a polynomial $P(x) \approx 1/x$ directly to the singular values of a *block-encoding* of A , reaching near-optimal $\tilde{O}(\kappa \log(1/\epsilon))$ and unifying HHL with Hamiltonian simulation (Section 6) and amplitude amplification (Section 5).
- *Provably optimal scaling.* Recasting $Ax = b$ as adiabatic ground-state preparation and invoking a discrete adiabatic theorem (Costa et al. 2022) achieves the optimal $O(\kappa \log(1/\epsilon))$ with explicit constants.
- *The classical counterpoint.* For *low-rank* systems, sample-and-query classical algorithms match the $\text{poly}(\log N)$ runtime (*dequantisation*; [5], Chia et al. 2020), so the exponential advantage in N is genuine only for sparse, high-rank, well-conditioned systems — exactly the regime carved out by the caveats above.

7.2 Caveats and honest assessment

The exponential speedup of HHL over classical solvers is conditional on four requirements, each of which is non-trivial.

1. **State preparation.** The input $|b\rangle$ must be prepared efficiently. Preparing an arbitrary N -dimensional state requires $O(N)$ gates in general, eliminating the $\log N$ advantage. QRAM (quantum random access memory) can prepare $|b\rangle$ in $O(\log N)$ time, but QRAM itself is a hardware resource that does not yet exist at scale.
2. **Sparse oracle.** HHL requires a sparse oracle for A : a quantum circuit that, given a row index, returns the non-zero entries. Constructing this oracle may itself require $O(N)$ resources.
3. **Condition number.** The runtime scales as $O(\kappa^2)$ (or $O(\kappa)$ with improved methods). For ill-conditioned systems ($\kappa \gg 1$) the speedup may be lost. Preconditioning is possible but requires quantum-compatible preconditioners.

4. **Output model.** HHL prepares $|x\rangle$, not the classical vector x . Extracting all N components of x from $|x\rangle$ requires $O(N)$ measurements (full tomography), again eliminating the speedup. HHL is useful only when one needs to compute a *functional* of x (e.g. $\langle x|M|x\rangle$ for some observable M), which can be estimated from $|x\rangle$ with $O(1)$ measurements.

Remark 7.7. The condition number κ plays exactly the same role in HHL as in classical iterative methods. For conjugate gradient, the iteration count is $O(\sqrt{\kappa} \log(1/\epsilon))$ and the total work is $O(N\sqrt{\kappa} \log(1/\epsilon))$ (one matrix–vector product per iteration). HHL’s total complexity $O(\log(N)\kappa^2/\epsilon)$ is better in N but worse in κ (by a factor $\kappa^{3/2}$) and in ϵ than CG. This means HHL outperforms CG only when N is enormous and κ is moderate. The post-selection probability $\Theta(1/\kappa^2)$ is a quantum analogue of the condition number penalty in classical solvers: a poorly conditioned system is hard to solve whether classically or quantumly, and preconditioning — reducing κ before solving — is as important in the quantum setting as in the classical one.

Example 7.8 (Discrete Poisson equation). The canonical application of HHL is the discrete Poisson problem $Lu = f$, with L the finite-difference Laplacian of Example 6.12 (Dirichlet boundary conditions, so that L is Hermitian positive definite). It is 3-sparse and $N = 2^n$ dimensional, and its evolution e^{iLt} is simulated in $\text{poly}(n)$ gates by the even-odd split-step there, so the QPE step of HHL is efficient and HHL prepares $|u\rangle \propto L^{-1}|f\rangle$.

It is also the cleanest illustration of the condition-number caveat. The Laplacian eigenvalues are $\lambda_k = 4 \sin^2(\frac{k\pi}{2(N+1)}) \in (0, 4)$, so $\lambda_{\max} = \Theta(1)$ while $\lambda_{\min} = \Theta(1/N^2)$, giving

$$\kappa(L) = \lambda_{\max}/\lambda_{\min} = \Theta(N^2).$$

The condition number grows polynomially in N — exponentially in n — so HHL’s $O(\log(N)\kappa^2) = O(nN^4)$ is far *worse* than the $O(N)$ classical solvers (tridiagonal elimination, multigrid, or the FFT): the exponential advantage in N is more than undone by the κ dependence. A genuine quantum speedup here would require preconditioning L to bounded κ and asking only for a scalar functional $\langle u|M|u\rangle$ of the solution, never the full grid function u .

Exercise 7.9. Consider a matrix A with eigenvalues uniformly spaced in $[1/\kappa, 1]$ and the constant $C = \lambda_{\min} = 1/\kappa$ in the controlled rotation. Show that for a uniform $|b\rangle$ (i.e. $\beta_j = 1/\sqrt{N}$ for all j) the post-selection probability is $\Theta(1/\kappa)$, whereas in the worst case ($|b\rangle$ aligned with the eigenvector of the largest eigenvalue) it is $\Theta(1/\kappa^2)$. The latter is the source of the κ^2 factor in HHL’s runtime.

Exercise 7.10. Explain why HHL does not give an exponential speedup for the problem of computing the full solution vector $x \in \mathbb{R}^N$. Under what conditions does HHL give a genuine advantage?